



EUROPEAN SPACE AGENCY

CALIBRATED DAILY FORECASTS OF THE F10.7 SOLAR RADIO FLUX TO 27 DAYS AHEAD

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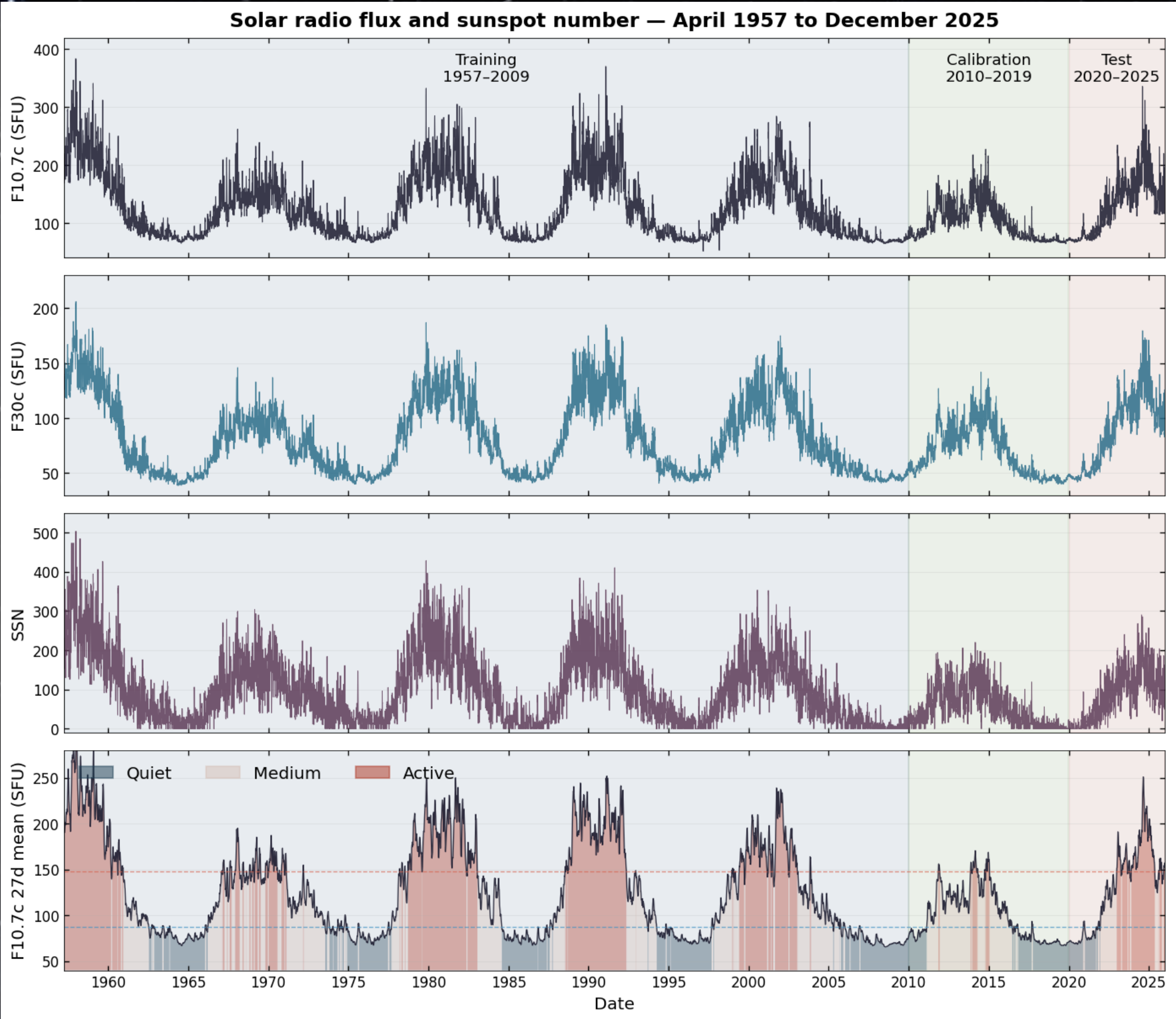
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Operational need → To improve the existing F10.7 forecast used for orbital predictions at ESOC.

- F10.7 is a standard solar proxy used in empirical thermosphere models.
- Forecast error propagates into density error and affects drag-sensitive LEO operations.
- The operational product must provide both a central forecast and a calibrated interval.

Methodology



Quiet < 86.8 SFU • Medium 86.8–147.4 SFU • Active > 147.4 SFU

For each horizon $h = 1 \dots 27$ days
Target: $\Delta h = F10.7c(t+h) - F10.7c(t)$

1 ElasticNet baseline

Regularised linear model with 5-fold time-series cross-validation. It captures the mostly linear recurrent structure.

2 HGBR residual correction

Gradient boosting is trained on ElasticNet residuals and adds the remaining nonlinear correction.

3 Weighted blend

ElasticNet and HGBR are combined by inverse out-of-fold RMSE weighting

4 Calibrated output

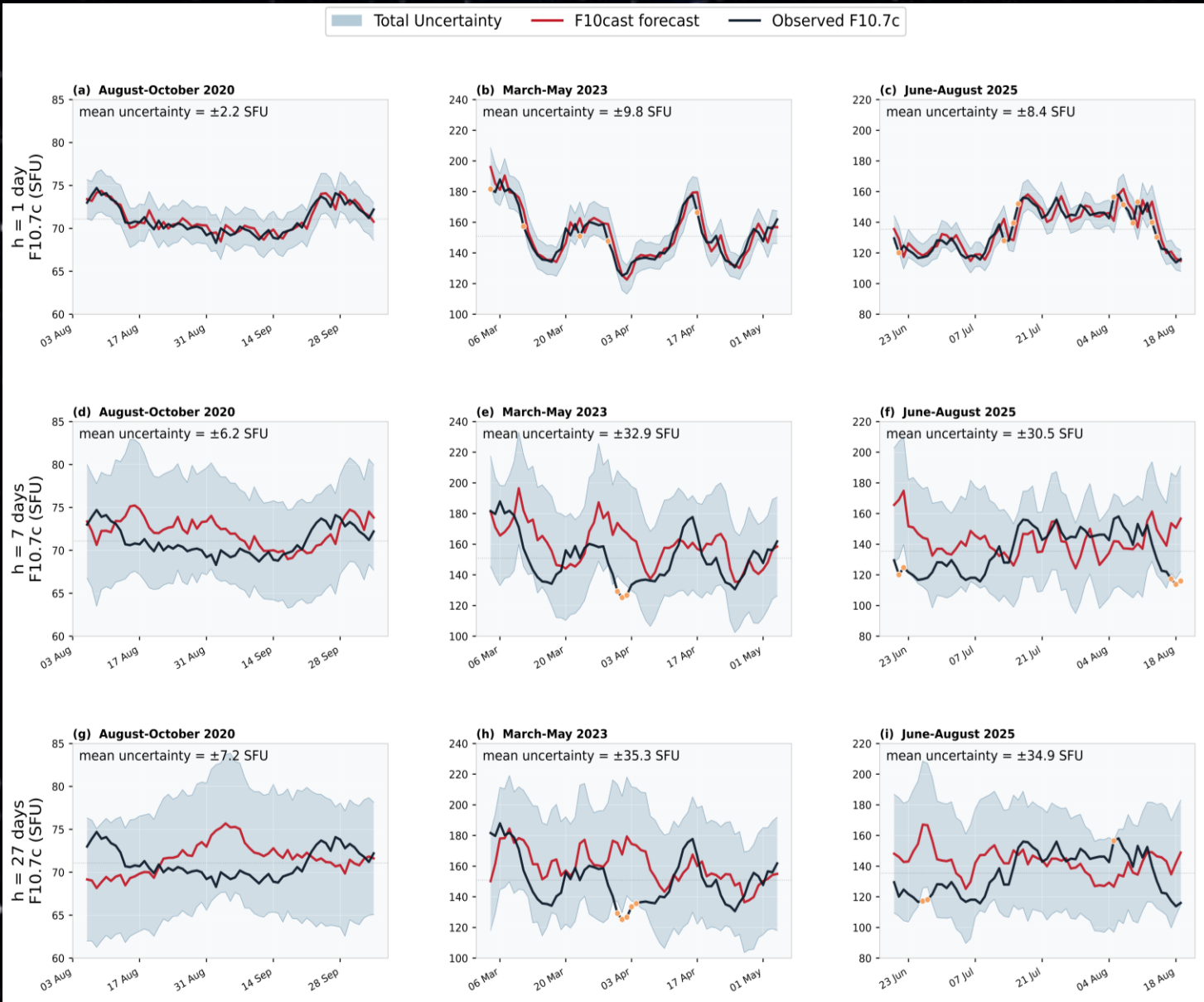
The point forecast is provided with a 90% interval from aleatoric, disagreement and conformal terms.

Operational output of F10Cast
daily F10.7c forecast + 90% prediction interval for $h = 1 \dots 27$ days

F10Cast Forecast and uncertainty

The interval is calibrated by horizon and activity.

Total uncertainty combines aleatoric spread, disagreement and a conformal calibration term.



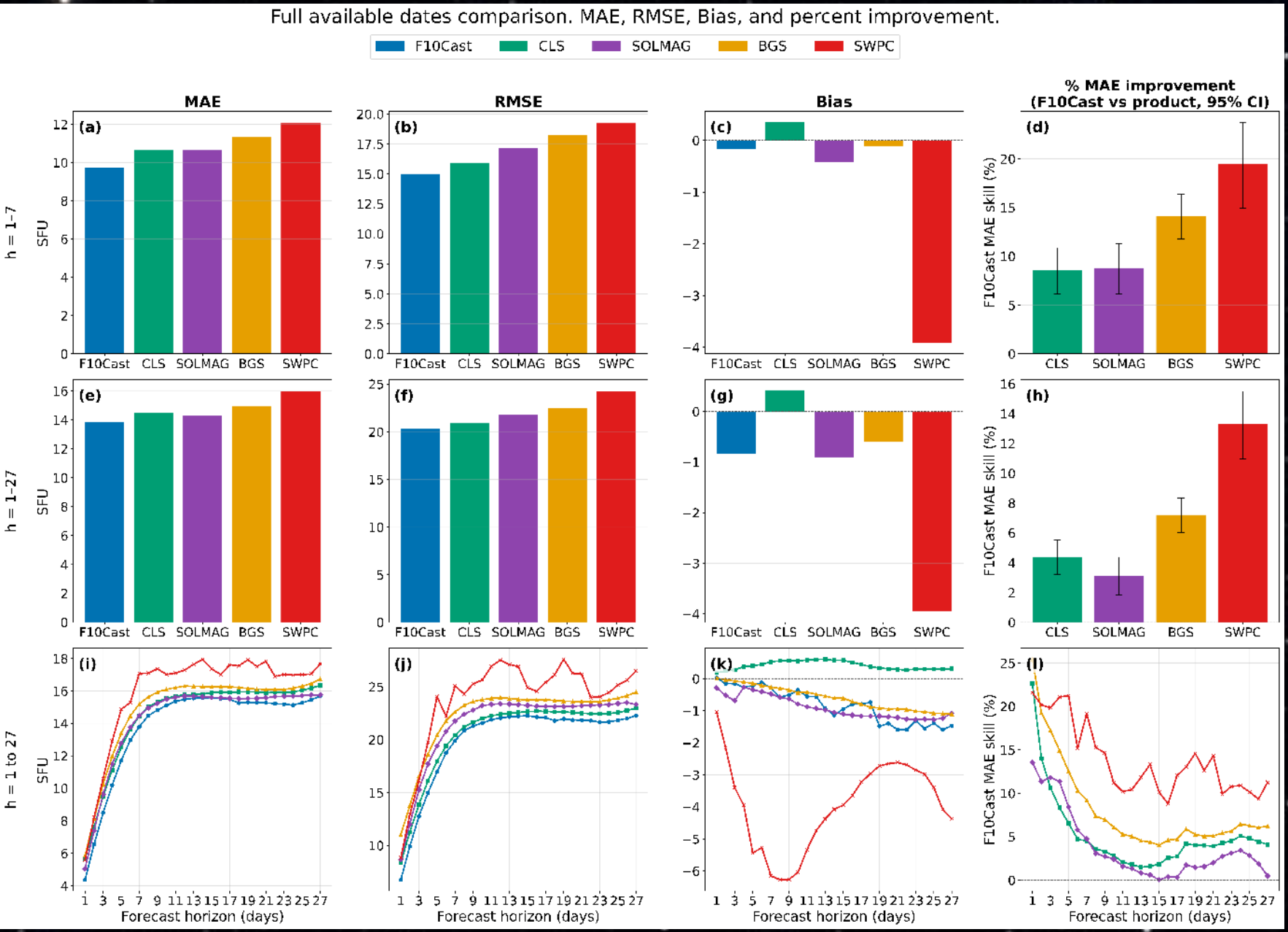
4.41 SFU
MAE at $h = 1$ day

14.11 SFU
MAE at $h = 7$ days

15.93 SFU
MAE at $h = 27$ days

Matched comparison with the 4 operational products

F10Cast, CLS, SOLMAG, BGS and NOAA/SWPC are compared on the same issue date, valid date and lead time.



MAE

–20%
vs SWPC

–14%
vs BGS

–10%
vs SOLMAG

–8%
vs CLS

Preliminary results:
Improved F10.7 skill carries through to lower thermospheric density error for altitudes 400 – 500 km. Density errors respond non-linearly to F10.7 forecasts.

